



Atomic superheterodyne receiver based on microwave-dressed Rydberg spectroscopy

Mingyong Jing^{1,2,3}, Ying Hu^{1,2,3}, Jie Ma^{1,2}, Hao Zhang^{1,2}, Linjie Zhang^{1,2} , Liantuan Xiao^{1,2} and Suotang Jia^{1,2}

Highly sensitive phase- and frequency-resolved detection of microwave electric fields is of central importance in a wide range of fields, including cosmology^{1,2}, meteorology³, communication⁴ and microwave quantum technology⁵. Atom-based electrometers^{6,7} promise traceable standards for microwave electrometry, but their best sensitivity is currently limited to a few $\mu\text{V cm}^{-1} \text{Hz}^{-1/2}$ (refs. ^{8,9}) and they only yield information about the field amplitude and polarization¹⁰. Here, we demonstrate a conceptually new microwave electric field sensor—the Rydberg-atom superheterodyne receiver (superhet). The sensitivity of this technique scales favourably, achieving even $55 \text{ nV cm}^{-1} \text{Hz}^{-1/2}$ with a modest set-up. The minimum detectable field of 780 pV cm^{-1} is three orders of magnitude smaller than what can be reached by existing atomic electrometers. The Rydberg-atom superhet allows SI-traceable measurements, reaching uncertainty levels of $10^{-8} \text{ V cm}^{-1}$ when measuring a sub- $\mu\text{V cm}^{-1}$ field, which has been inaccessible so far with atomic sensors. Our method also enables phase and frequency detection. In sensing Doppler frequencies, sub- μHz precision is reached for fields of a few hundred nV cm^{-1} . This work is a first step towards realizing electromagnetic-wave quantum sensors with quantum projection noise-limited sensitivity. Such a device will impact diverse areas like radio astronomy, radar technology and metrology.

Quantum sensing harnesses highly coherent and well-controlled quantum systems to measure weak signals with unprecedented sensitivity and precision¹¹. In particular, the Rydberg atom provides a platform that allows high-sensitivity detection of a microwave (MW) electric field^{6,11,12}. This has been highlighted by recent experiments, with the development of traceable Rydberg-atom electrometers that outperform their classical counterparts^{8–10,13–15} and sensing techniques with subwavelength spatial resolution enabled by optical readout^{16,17}.

Two central quests in sensing time-varying electric fields are (1) to diagnose the phase and frequency of weak fields, which is at the basis of (as a paradigmatic example) radar detections¹⁸ and (2) to improve sensitivity. However, existing atom-based MW field sensors only allow us to probe the field amplitude and polarization vector from the optical readout^{7–10,13–17,19}, severely limiting their practical applications in actual MW detections. Proposals to measure the phase and frequency—involving complicated techniques—have been reported^{6,20}, but demonstrations with experimentally realistic tools have remained elusive. Furthermore, the sensitivity of atom-based electrometers has approached the photon shot noise limit, with $\sim 2 \mu\text{V cm}^{-1} \text{Hz}^{-1/2}$ at best, even when assisted by

sophisticated approaches⁹. Beating this limit is very demanding as it requires non-classical squeezed light in relevant atomic set-ups. The origin for such difficulties in improving the sensitivity of existing atom-based electrometers is that they only realize nonlinear detection when the signal is so weak that the corresponding Rabi frequency is smaller than the linewidth of the optical spectrum, leading to an (approximately) unfavourable scaling of sensitivity ($\propto \sigma^{1/2}$) with classical noise-induced error σ . An alternative route to high sensitivity exploits non-classical Schrödinger cat states, as recently demonstrated²¹, but the experiment is infeasible in all but very small atomic systems due to the difficulty in preparing cat states. Developing new sensing techniques that overcome both above limitations is currently an outstanding challenge.

Here, we realize a conceptually new MW electric-field sensor based on MW-dressed Rydberg atoms²² and a tailored electromagnetically induced transparency (EIT) spectrum^{23,24}. Not only can it resolve the phase and frequency of MWs, but it also realizes a very favourable $\propto \sigma$ scaling of sensitivity, providing a remarkably high sensitivity of $\lesssim 55 \text{ nV cm}^{-1} \text{Hz}^{-1/2}$ that is accessible to experiments even without non-classical resources. This Rydberg-atom sensor is functionally reminiscent of an electronic-circuit-based superheterodyne receiver (superhet) in conventional electromagnetic-wave detection architectures^{4,18}, but nonetheless has important advantages enabling traceable electric field measurements with high accuracy.

We present a Rydberg-atom superhet based on the set-up shown in Fig. 1a. The key new ingredient is a strong local MW field resonant with Rydberg transitions, with a Rabi frequency Ω_L . Our goal is to measure a signal MW that leads to a coupling $\Omega_s e^{-i(2\pi\delta_s t + \phi_s)}$ between Rydberg states with $\Omega_s \ll \Omega_L$, and both frequency detuning δ_s and phase ϕ_s are measured relative to the local field. We consider $2\pi\delta_s \ll \Gamma_{\text{EIT}}$ where Γ_{EIT} is the typical EIT linewidth.

In an atomic superhet, the entire information in a MW signal is encoded in the first-order energy shift of MW-dressed Rydberg states. The strong on-resonant local MW field results in two dressed states $|\pm\rangle$ energetically separated by $\hbar\Omega_L$, where $\hbar$ is the Planck constant; these are, respectively, the symmetric and antisymmetric superpositions of two bare Rydberg states $|3\rangle$ and $|4\rangle$. When perturbed by a small MW signal, $|\pm\rangle$ acquires an instantaneous energy shift $\pm E_1 = \pm \hbar\Omega_s \cos(2\pi\delta_s t + \phi_s)/2$.

In detecting E_1 , we use Ω_L as a control knob to engineer the EIT spectrum of the Rydberg atoms, as schematically shown in Fig. 1b. Because the strong local MW induces Autler–Townes (AT) splitting²³ of the EIT peak, the position of the on-resonance transmission point $\bar{P}_0$ is moved to the hillside of each EIT line, which has a slope of magnitude $|k|$. Consequently, when energy perturbations

¹State Key Laboratory of Quantum Optics and Quantum Optics Devices, Institute of Laser Spectroscopy, Shanxi University, Taiyuan, Shanxi, China.

²Collaborative Innovation Center of Extreme Optics, Shanxi University, Taiyuan, Shanxi, China. ³These authors contributed equally: Mingyong Jing, Ying Hu.

e-mail: zlj@sxu.edu.cn; xlt@sxu.edu.cn

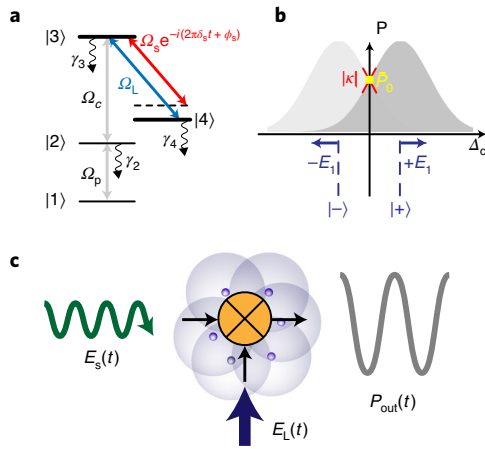


Fig. 1 | Basic principle of the Rydberg-atom superhet. **a**, Level scheme. States $|1\rangle$, $|2\rangle$ and Rydberg state $|3\rangle$ are resonantly coupled by probe (Ω_p) and control (Ω_c) fields, respectively. A local MW electric field (blue) resonantly drives Rydberg transition $|3\rangle - |4\rangle$ with Rabi frequency Ω_L . A weak signal MW (red) yields a coupling $\Omega_s e^{-i(2\pi\delta_s t + \phi_s)}$ with phase ϕ_s and frequency δ_s relative to the local MW. State $|i\rangle$ ($i = 2, 3, 4$) has decay rate γ_i . **b**, Schematic of MW sensing with MW-dressed Rydberg EIT spectroscopy. Owing to the strong local MW, probe laser transmission P as a function of control laser detuning Δ_c exhibits AT splitting of the EIT peaks. Resonant transmission point P_0 sits on the hillsides (with slope $|\kappa|$) of the EIT lines, where small energy shifts $\pm E_1 = \pm \hbar\Omega_s \cos(2\pi\delta_s t + \phi_s)/2$ in the dressed Rydberg states $|\pm\rangle$ give linear changes in transmission. **c**, An atomic superhet consisting of Rydberg atoms dressed by a local MW $E_L(t)$ detects a MW signal $E_s(t)$ as an optical output $P_{out}(t)$ that has downshifted frequency, original phase and linearly magnified amplitude.

$\pm E_1$ shift both EIT lines outwards, this shift transfers linearly—at a rate κ —into changes in the resonant light transmission. By tuning Ω_L , we can tailor the profile of the EIT lines, thereby effectively controlling the position of $\bar{P}_0$ therein (Fig. 1b). This allows us to achieve an optimal point that has maximum slope $|\kappa_0|$ when $\bar{P}_0$ sits around the half-maximum of both EIT lines.

The Rydberg-atom superhet thus detects a MW signal as an oscillating optical readout (Fig. 1c), represented by the change $P_{out}(t) = P(t) - \bar{P}_0$ in the resonant light transmission with and without signals. When operated at the optimal point, it yields (see Methods)

$$P_{out}(t) = |P(\delta_s)| \cos(2\pi\delta_s t + \phi_s) \quad (1)$$

Here, $|P(f)|$ is the amplitude of the single-sided Fourier spectrum of $P(t)$ at linear frequency f , and the Rabi frequency Ω_s of the signal is measured as

$$\Omega_s = \frac{|P(\delta_s)|}{|\kappa_0|} \quad (2)$$

This provides us with a signal amplitude $E_s = (1/\sqrt{2})\hbar\Omega_s/\mu$, where μ is the dipole moment associated with the Rydberg transition. Note that the root-mean-square (r.m.s.) amplitude of the electric field is used throughout this work. Crucially, both $|\kappa_0|$ and $|P(\delta_s)|$ are directly derived from the optical spectrum (see Methods), thereby essentially reducing the electric-field measurement to an optical-frequency measurement and ensuring SI-traceability of the measurement.

Equations (1) and (2) are at the heart of atomic superhet-based MW sensing. It offers significant advantages over existing atomic

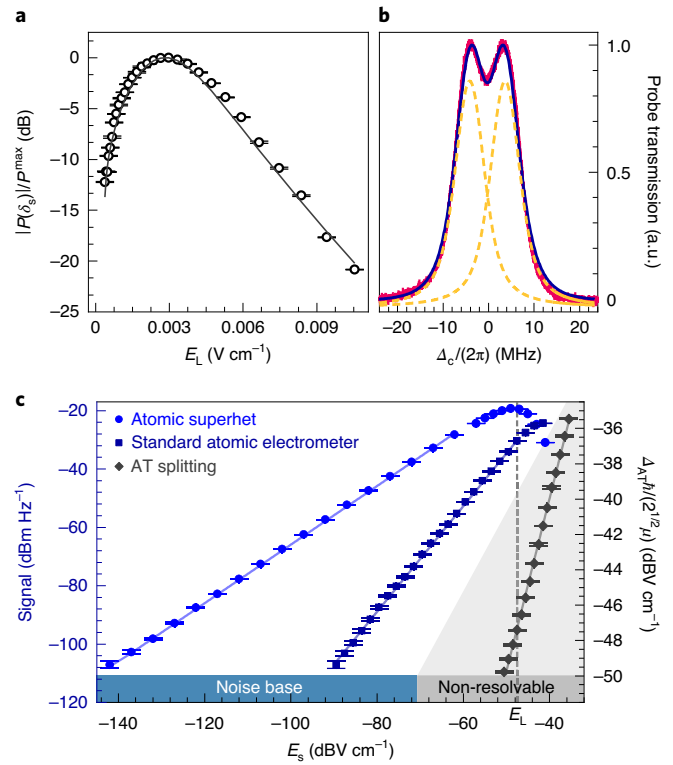


Fig. 2 | Proof-of-principle demonstrations of the optimal point and linear detection. **a**, $|P(\delta_s)|/P^{\max}$ as a function of E_L for a test signal, where $P^{\max}$ is the maximum of $|P(\delta_s)|$. Experimental data are shown as black circles. Error bars denote 1σ standard error obtained from 10 measurements. $P^{\max}$ occurs here for $E_L = 3.0 \text{ mV cm}^{-1}$ (r.m.s.), corresponding to $\Omega_L/(2\pi) = 7.9 \text{ MHz}$. The solid grey curve shows theoretical density-matrix calculations. **b**, Probe laser transmission as a function of control laser detuning Δ_c , corresponding to the peak in **a**. The red curve depicts the data recorded by a digital acquisition oscilloscope. The solid blue curve shows density-matrix calculations. Dashed yellow curves show a two-peak fit of experimental data using Voigt functions. **c**, Linear versus nonlinear detections. For the atomic superhet, $|P(\delta_s)|$ is measured (blue circles) for various values of signal field E_s . Error bars denote 1σ standard error from 10 measurements. The linear dynamical range of the atomic superhet is 90 dB. $E_L = 3.0 \text{ mV cm}^{-1}$ is indicated by a dashed grey line. For comparison, we converted our set-up to the standard configuration of atomic electrometers^{6,7}, and the difference $\Delta P(E_s) = P(E_s) - P(0)$ in the resonant probe transmission (dark blue squares) with and without signal was measured. Error bars indicate 1σ standard error from 10 measurements. Note: to avoid low-frequency noises for atomic electrometers, a 100% amplitude modulation at 150 kHz was added to the signal and demodulated with a spectrum analyser with 1 Hz resolution bandwidth (RBW). The blue region denotes the optical readout noise base. For strong E_s we apply a standard AT-splitting (Δ_{AT}) measurement technique to obtain a field value $\hbar\Delta_{AT}/(\sqrt{2}\mu)$ (grey diamonds). Error bars denote 1σ standard error from three measurements. The grey region corresponds to experimentally non-resolvable AT splittings. All solid curves in **c** are linear fits to the data. In **a** and **c**, $\delta_s = 150 \text{ kHz}$ and $|P(\delta_s)|$ is recorded with a spectrum analyser with 1 Hz RBW. E_L in **a** and E_s in **c** are calibrated using the standard antenna method; these are assumed to be the true values here.

MW sensors in the measurement of ultra-weak electric fields in the regime $\Omega_s \ll \Gamma_{\text{EIT}}$: as well as phase and frequency resolution, it affords a substantial sensitivity gain in traceable amplitude measurements. In particular, the atomic superhet has favourable scalings of sensitivity ($S \propto \sigma$ with classical noise-induced error σ) and the minimum detectable field ($E_{\min} = S/\sqrt{T}$ with measurement

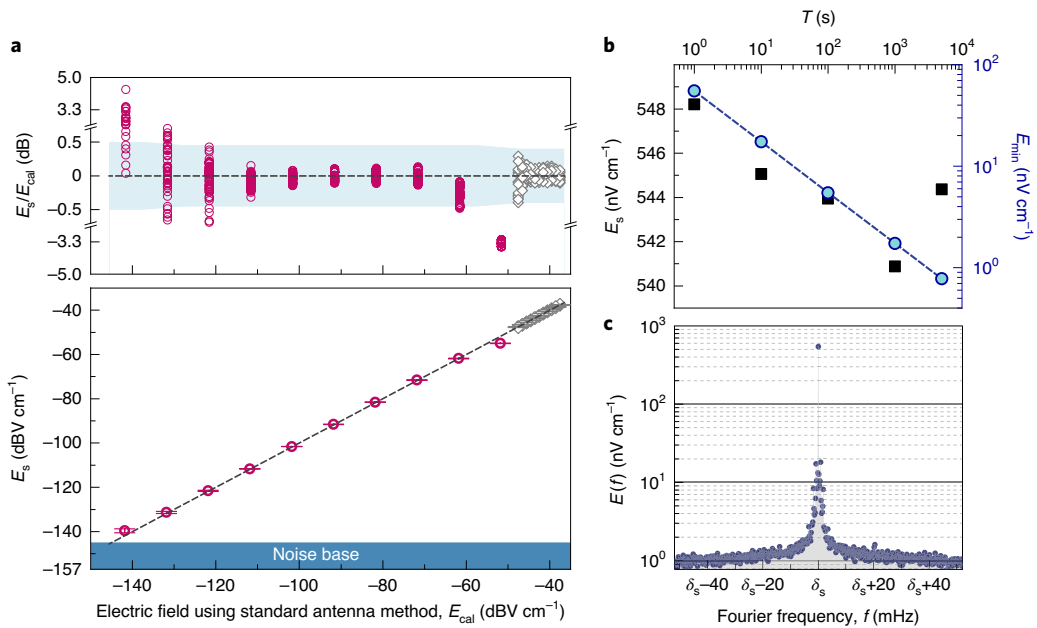


Fig. 3 | Sensitivity and uncertainties of the electric field measurement. **a**, Electric field E_s , measured by the atomic superhet, compared with electric field E_{cal} measured by the standard antenna method. Dashed lines in the top and bottom panels correspond to average values of E_{cal} . The light blue shaded area in the top panel depicts measurement uncertainties in the standard antenna method (see Methods). Red circles in the top panel represent single-shot measurements by the atomic superhet in a 1 s measurement time. For every field, 50 single-shot measurements were performed over five days (10 per day). Their averages are shown by corresponding red circles in the bottom panel. Error bars indicate 1σ standard error. In strong-field regimes, for each signal we performed 15 single-shot AT-splitting measurements (grey diamonds, top panel). Their averages are denoted by the corresponding grey diamonds in the bottom panel. Error bars indicate 1σ standard error. The blue region in the bottom panel denotes the noise base of the electric field measured by the atomic superhet in a 1 s measurement time, indicating a sensitivity of $S = 55 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$. **b**, Measurements of the same MW signal by the atomic superhet at different measurement times T . Black squares denote average E_s values obtained from 50 single-shot measurements at T , which show no visible systematic error up to $T = 5,000$ s. Blue circles depict the smallest detectable field E_{min} at T obtained from the optical readout noise base at T divided by κ_0 . The dashed line shows the calculated E_{min} using the characteristic relation $E_{\text{min}} = S/\sqrt{T}$ of slope detection. **c**, A single-sided Fourier spectrum of MW field $E(f) = |\hbar/(\sqrt{2}\mu)|P(f)|/\kappa_0|$ measured at $T = 5,000$ s. From the peak we derive a SNR of 57 dB, agreeing with the SNR predicted from the preset electric field value via the standard antenna method. In all panels, $\delta_s = 150$ kHz.

time T). This is in contrast to standard atomic electrometers^{6,7}, which for $\Omega_s \ll \Gamma_{\text{EIT}}$ realize nonlinear detection, where $S \propto \sqrt{\sigma}$ and $E_{\text{min}} = S/\sqrt{T}$. Combined with operation at the optimal point, the atomic superhet maximizes efficiencies in the effort to enhance sensitivity by reducing classical noise or to access smaller fields by increasing the measurement time.

We implemented our Rydberg-atom superhet using Cs atoms in a room-temperature vapour cell (see Methods). Figure 2 provides experimental verifications for the optimal point and linear relation. $|P(\delta_s)|$, scanned as a function of the local MW field, features a maximum (Fig. 2a), indicating the existence of an optimal point. The EIT spectrum corresponding to this maximum (Fig. 2b), combined with a double-peak fit, shows that the resonant transmission point therein indeed sits at the half-maximum of each EIT line or, equivalently, $\Omega_L \sim \Gamma_{\text{EIT}}$. The solid curves in Fig. 2a,b represent our theoretical calculations (see Methods), which agree well with the experiment. Figure 2c corroborates $|P(\delta_s)| \propto E_s$ for the atomic superhet when $\Omega_s \ll \Omega_L$, but the linearity begins to break down when $\Omega_s \sim \Omega_L$. To further illustrate the advantage of linear over nonlinear detections in the regimes $\Omega_s \ll \Gamma_{\text{EIT}} \sim \Omega_L$ given the same optical readout noises, we converted our set-up into the standard configuration of atomic electrometer from refs. ^{6,7}. Its optical output (Fig. 2c) was then found to scale as $\propto E_s^{1.8}$ (that is, nearly quadratic) with E_s , which reaches unit signal-to-noise ratio (SNR) at a substantially larger E_s than the atomic superhet. Note that, for $\Omega_s > \Gamma_{\text{EIT}}$, a linear detection of E_s is viable through an AT-splitting measurement.

Having established our sensing principle, we now demonstrate the outstanding performance of the atomic superhet in MW

measurements, based on equations (1) and (2). From the electric field E_s obtained in a measurement time of 1 s (Fig. 3a), we derive a sensitivity of $S = 55 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$, which represents an improvement by a factor of 38 over the state of the art^{8,9}. To find the smallest detectable field E_{min} that can be achieved, we reduce the technical noise affecting the long-time stability of our system (see Methods). Measurements of the same signal at different times display a random fluctuation within 1% up to $T = 5,000$ s (Fig. 3b), showing the stability of our system over this time. The measured $E_{\text{min}}(T)$ agrees with $E_{\text{min}} = S/\sqrt{T}$, as expected. We thus achieve $E_{\text{min}} = 780 \text{ pV cm}^{-1}$ for $T = 5,000$ s. Figure 3c presents a typical (single-sided) Fourier spectrum of the MW electric field at $T = 5,000$ s. Because T is, in principle, limited by the stability of the external synchronization clock and can be extended to more than 4 h (ref. ²⁵), we expect to reach $E_{\text{min}} \lesssim 460 \text{ pV cm}^{-1}$. Figure 3a shows that the atomic superhet also achieves a higher performance than all the other atomic sensors in the accuracy of the measurement of a MW field below $8 \mu\text{V cm}^{-1}$. The smallest uncertainty is at levels of $10^{-8} \text{ V cm}^{-1}$ for 1 s measurement time, which is essentially determined by the precision of the optical frequency measurement based on the EIT spectrum (see Methods for the uncertainty analysis). Figure 3a, moreover, compares measurements by the atomic superhet and by a standard antenna method (see Methods). They agree, within their combined uncertainty of $\sim 5\%$, for $177 \text{ nV cm}^{-1} \leq E_s \leq 1.78 \text{ mV cm}^{-1}$. We believe that this uncertainty originates from that associated with the antenna approach in setting the output power of the MW generator. The relative statistical uncertainty in a single-shot measurement by the atomic superhet increases with a reduced signal,

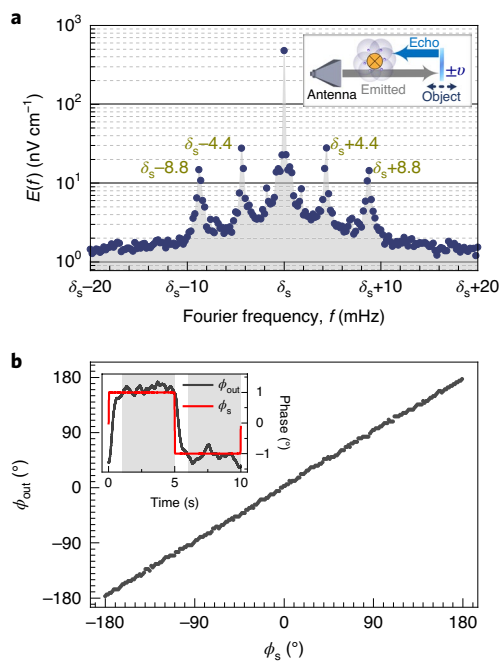


Fig. 4 | Frequency and phase detections. **a**, Doppler frequency measurement. Inset: an emitted MW, with frequency detuning $\delta_s = 150$ kHz from the local MW (6.947 GHz), is reflected from a target moving periodically relative to the antenna. The atomic superhet receives both emitted and echo MWs to produce an optical output $P(t)$. The single-sided Fourier spectrum, $E(f)$, of the electric field is obtained from an average of 50 continuous measurements of $P(t)$, each sampled over a 5,000 s measurement time. The horizontal axis indicates detuning of the Fourier frequency from the local MW. The preset radial speed of the object, determined by the linear stage, is $100 \mu\text{m s}^{-1}$ and the periodicity is 6,000 s. **b**, Comparisons between the measured phase ϕ_{out} and original phase ϕ_s in the signal, $\Omega_s e^{-i(2\pi\delta_s t + \phi_s)}$. In the experiment, $\Omega_s/(2\pi) = 20$ kHz and $\delta_s = 150$ kHz. ϕ_{out} was obtained from phase-sensitive measurements of $P(t)$. Inset: ϕ_{out} for a phase ϕ_s switching between $\pm 1^\circ$. The standard error of ϕ_{out} was obtained for the data sampled in the time interval in the grey shaded block (see Methods).

but can be decreased by averaging over multiple measurements. Systematic uncertainty for the atomic superhet only becomes significant when $E_s < 177 \text{ nV cm}^{-1}$ and $E_s > 1.78 \text{ mV cm}^{-1}$, the former due to the incoherent sum of signal and noise and the latter due to breakdown of the linear relation. Our measurements in ultrasmall field regimes complement the AT-splitting measurement for strong fields, $E_s \gtrsim 4.0 \text{ mV cm}^{-1}$.

Figure 4a showcases the capability of the atomic superhet to sense the Doppler frequency shift and discern the motion of targets, which has important roles in fields such as aviation, meteorology and biomedical science. In our experiment, an emitted MW is reflected from a periodically moving object, which is received (along with the emitted MW) by the atomic superhet to produce $P_{\text{out}}(t)$, simulating Doppler radar detection¹⁸. The corresponding single-sided Fourier amplitude spectrum displays five pronounced peaks (Fig. 4a), the central one corresponding to the emitted frequency. The measured Doppler frequency shift, $\delta f = \pm 4.4$ mHz, allows us to determine the radial speed of the object toward (or away from) the antenna as $v = 95 \mu\text{m s}^{-1}$. The two side peaks, $2\delta f$ away from the centre, indicate the occurrence of multiple reflections. The frequency resolution is 0.2 mHz, limited by the resolution bandwidth (RBW) of the fast Fourier transform (FFT). The frequency precision, $\text{RBW}/\sqrt{2} \times \text{SNR}$, reaches sub- μHz for fields of

hundreds of nV cm^{-1} . The bandwidth of the current atomic superhet is in excess of 100 kHz, which allows us to measure velocities from a few $\mu\text{m s}^{-1}$ to a few km s^{-1} , covering motions from cell migration to the rapid flight of rockets.

The ability of the atomic superhet to resolve the phase of the MW is demonstrated in Fig. 4b. The phase ϕ_{out} extracted from $P(t)$ (see Methods) and the true phase ϕ_s show good agreement. By measuring the ϕ_{out} that corresponds to a jumping input phase ϕ_s (inset), we estimate a phase resolution of 0.8° , consistent with theory (see Methods).

In conclusion, we have developed a new MW sensing technique based on MW-dressed Rydberg spectroscopy, which simultaneously allows for SI-traceable measurement of a signal electric field with ultrahigh sensitivity as well as the detection of phase and frequency. By producing favourable scalings, our technique has significantly improved the efficiencies in ongoing experimental efforts to reduce classical noise for sensitivity gain, thus opening a feasible route towards a quantum projection noise-limited (QPNL) sensitivity of $\sim 700 \text{ pV cm}^{-1} \text{ Hz}^{-1/2}$ (see Methods), which is challenging in current experiments. A further boost of sensitivity is anticipated from combinations of the present technique and non-classical resources. Equally appealing is the prospect of applying the Rydberg-atom superhet to traceable MW electric-field standards in ultrasmall-field regimes. This technique can be generalized to detect electromagnetic radiation from the radiofrequency regime to the far-infrared and is feasible for miniaturization^{26–28}. Our work provides a remarkable step on the path to realizing future quantum receivers, such as in radars or radio telescopes, with the benefits of SI-traceable accuracy and ultrahigh sensitivity, or in terahertz communication^{29,30}, allowing us to recover information encoded in terahertz carriers via phase or frequency modulation.

Online content

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Methods

Theory. Here, we derive the probe laser transmission for the Rydberg-atom superhet based on the set-up in Fig. 1a. Under normal resonant conditions, $\Delta_{\text{pic}} = 0$ for coupling and probe lasers, and the relevant Hamiltonian takes the form (in the basis of bare states $[|1\rangle, |2\rangle, |3\rangle, |4\rangle]^T$)

$$H(t) = \hbar \begin{pmatrix} 0 & \frac{\Omega_p}{2} & 0 & 0 \\ \frac{\Omega_p}{2} & 0 & \frac{\Omega_c}{2} & 0 \\ 0 & \frac{\Omega_c}{2} & 0 & \frac{\Omega_L + e^{-iS(t)}\Omega_s}{2} \\ 0 & 0 & \frac{\Omega_L + e^{iS(t)}\Omega_s}{2} & 0 \end{pmatrix} \quad (3)$$

where $S(t)$ denotes the time-dependent relative phase $S(t) = 2\pi\delta_s t + \phi_s$ between the signal and local MW.

Accounting for spontaneous emission, the dynamics of our system is governed by the following master equation for the (four-level) density matrix ρ :

$$\dot{\rho} = \frac{i}{\hbar} [\rho, H(t)] + \mathcal{D}[\rho] \quad (4)$$

where the second term is given by

$$\mathcal{D}[\rho] = \begin{pmatrix} \gamma_2\rho_{22} + \gamma_4\rho_{44} & -\frac{\gamma_2}{2}\rho_{12} & -\frac{\gamma_3}{2}\rho_{13} & -\frac{\gamma_4}{2}\rho_{14} \\ -\frac{\gamma_2}{2}\rho_{21} & \gamma_3\rho_{33} - \gamma_2\rho_{22} & -\frac{\gamma_3}{2}\rho_{23} & -\frac{\gamma_4}{2}\rho_{24} \\ -\frac{\gamma_3}{2}\rho_{31} & -\frac{\gamma_3}{2}\rho_{32} & \gamma_4\rho_{44} - \gamma_3\rho_{33} & -\frac{\gamma_4}{2}\rho_{34} \\ -\frac{\gamma_4}{2}\rho_{41} & -\frac{\gamma_4}{2}\rho_{42} & -\frac{\gamma_4}{2}\rho_{43} & -\gamma_4\rho_{44} \end{pmatrix}$$

Here, $\gamma_i = (\gamma_i + \gamma_i)$, where γ_i ($i = 2, 3, 4$) is the decay rate (Fig. 1a). Note that we have ignored the spontaneous emission associated with $|3\rangle \rightarrow |4\rangle$ and other possible transitions, as they are comparatively small. We have also ignored the laser linewidths, which, as shown below, are a few tens of Hz in our experiment, much smaller than the relevant coherence and spontaneous decay rates. Our goal is to derive the probe laser transmission in the adiabatic limit of equation (4) when δ_s is small compared to the characteristic energy scales of the system dynamics.

To gain an intuitive understanding, we begin with analytical derivations based on cold atoms. Within the adiabatic approximation, the probe laser transmission associated with the instantaneous steady state is written in terms of the imaginary component of the susceptibility as²³ $P(t) = P_i e^{-kL\chi(t)}$. Here, P_i is the incident light power, L is the length of the cell containing atoms and $k = 2\pi/\lambda_p$ is the wavevector of the probe laser. Note that $\chi(t) = C\rho_{21}(t)$ is the susceptibility, where $\rho_{21}(t)$ denotes the instantaneous steady-state density matrix component associated with the $|1\rangle \rightarrow |2\rangle$ transition. Furthermore, $C = -2N_0\mu_{12}/(\epsilon_0\hbar\Omega_p)$, where N_0 is the total density of atoms, μ_{12} is the dipole moment of the ground-state transition and ϵ_0 is the permittivity in vacuum. When $\Omega_s \ll \Omega_L$, assuming $\gamma_{3(4)} = 0$, the imaginary part of $\chi(t)$ under normal resonant conditions is derived as

$$\Im[\chi(\Omega, t)] = \chi_0 \frac{|\Omega|^2}{|\Omega|^2 + \Gamma^2} \quad (5)$$

where

$$\Gamma = \Omega_p \sqrt{\frac{2(\Omega_c^2 + \Omega_p^2)}{2\Omega_p^2 + \gamma_2^2}}$$

is intimately related to the EIT linewidth,

$$\chi_0 = \frac{C\gamma_2\Omega_p}{\gamma_2^2 + 2\Omega_p^2}$$

is the peak value of the spectrum, and $\Omega = |\Omega_L + e^{-iS(t)}\Omega_s|$. Perturbative expansions of equation (5) in terms of the small parameter Ω_s/Ω_L at first order give $\Im[\chi(\Omega_L, t)] = \Im[\chi(\Omega_L)] + S_L\Omega_s \cos(2\pi\delta_s t + \phi_s)$. Here

$$S_L = 2\chi_0 \left[\frac{\Gamma^2\Omega_L}{(\Omega_L^2 + \Gamma^2)^2} \right]$$

is the slope of function (5) at $\Omega = \Omega_L$. When $\Omega_L = \Gamma/\sqrt{3}$, the spectrum is linear near $\Omega = \Omega_L$, corresponding to the maximum slope $S_{\text{max}} = 3\sqrt{3}\chi_0/(8\Gamma)$. Denoting $\alpha = kLS_{\text{max}}$, for $\alpha\Omega_s/\Gamma \ll 1$, we find the transmission as $P_t = P_0 + (\alpha P_0/\Gamma)\Omega_s \cos(2\pi\delta_s t + \phi_s)$, with $P_0 = P_i e^{-kL\chi_0/4}$.

In general, for $\Omega_s \ll \Omega_L$, a linear expansion of the resonant transmission around P_0 generically leads to the expression

$$P(t) = \bar{P}_0 + \kappa\Omega_s \cos(2\pi\delta_s t + \phi_s) \quad (6)$$

where κ is the linear expansion coefficient defined by $\kappa = \partial\bar{P}_0/\partial\Omega_L$. A key point to stress here is that κ can be obtained from the EIT spectrum $\bar{P}(\Delta_c)$ for $\Omega_s = 0$, where Δ_c is the control laser detuning. (Note that, theoretically, $\bar{P}(\Delta_c)$ can be calculated in a standard way based on the four-level model in the presence of $\Delta_c \neq 0$.) We have

$\bar{P}_0 = \bar{P}(\Delta_c)|_{\Delta_c=0}$. Due to AT splitting induced by the strong Ω_L , the spectrum $\bar{P}(\Delta_c)$ can be described in terms of a superposition of two (three-level) EIT lines represented by $\mathcal{F}_{\pm}(\Delta_c - \Omega_{\pm})$ peaking at $\Omega_{\pm}$, respectively. Here, $\Omega_{\pm}$ correspond to energies of the dressed Rydberg states $|\pm\rangle$ (Figs. 1b and 2b), with $\Omega_+ - \Omega_- = \Omega_L$. That is

$$\bar{P}(\Delta_c) = \mathcal{F}_+(\Delta_c - \Omega_+) + \mathcal{F}_-(\Delta_c - \Omega_-) \quad (7)$$

When all lasers are locked on resonance with the relevant atomic transitions, we have $\partial_{\Delta_c}\mathcal{F}_+|_{\Delta_c=0} = -\partial_{\Delta_c}\mathcal{F}_-|_{\Delta_c=0}$ and obtain

$$\kappa \equiv \frac{\partial\bar{P}(\Delta_c)}{\partial\Omega_L}|_{\Delta_c=0} = \frac{1}{2} \left(\frac{\partial\mathcal{F}_-}{\partial\Delta_c}|_{\Delta_c=0} - \frac{\partial\mathcal{F}_+}{\partial\Delta_c}|_{\Delta_c=0} \right) \quad (8)$$

The optimal point is a special point for which the slope amplitude $|\kappa_0|$ is maximum; that is, it satisfies the condition

$$\frac{\partial\kappa(\Omega_L)}{\partial\Omega_L} = \frac{\partial^2\bar{P}_0(\Omega_L)}{\partial\Omega_L^2} = 0 \quad (9)$$

Operating the sensor at this optimal point brings an additional advantage: by definition, fluctuations in Ω_L will not affect the value of κ_0 , at least at first order. Motivated by previous analysis for cold atoms, we formally write

$$\kappa_0 = \frac{\alpha\bar{P}_0}{\Gamma_{\text{sp}}} \quad (10)$$

Here, α denotes the fraction of photons participating in the EIT, Γ_{sp} is intimately related to the typical EIT linewidth Γ_{EIT} , and we interpret Γ_{sp}^{-1} as the coherence time of the Rydberg-atom superhet.

Equations (6), (8) and (9) provide the key equations of our sensing principle for high-sensitivity traceable measurement of Ω_s (see equation (2)). Finally, we comment on the Doppler effect. For the transmission spectrum obtained by scanning the control laser, as in the present case, a correction for Doppler mismatch²⁴ is not needed²⁴. However, the Doppler broadening effect will affect the EIT profile and thus the value of κ_0 , although it will not change the form of equations (6) to (9). Theoretically, the Doppler broadening effect is accounted for through the standard Doppler average of (instantaneous) steady-state ρ_{21} before calculating the transmission, as in our theoretical calculations, shown in Fig. 2a,b.

Experimental set-up. For the experiment, we used Cs atoms in a vapour cell at room temperature. The cell was 5 cm long and contained ground-state atoms at a total density of $N_0 = 4.89 \times 10^{10} \text{ cm}^{-3}$. We realized the four-level configuration in Fig. 1 using four states in a Cs atom: $6S_{1/2}$, $F=4$; $6P_{3/2}$, $F=5$; $47D_{5/2}$; $48P_{3/2}$. The hyperfine states $6S_{1/2}$, $F=4$ and $6P_{3/2}$, $F=5$ comprised the lowest two states $|1\rangle$ and $|2\rangle$ in the configuration, with $\gamma_2/(2\pi) = 5.2 \text{ MHz}$. The Rydberg state $47D_{5/2}$, with inverse lifetime $\gamma_3/(2\pi) = 3.9 \text{ kHz}$, and Rydberg state $48P_{3/2}$, with inverse lifetime $\gamma_4/(2\pi) = 1.7 \text{ kHz}$, made up states $|3\rangle$ and $|4\rangle$. In calculating γ_3 and γ_4 at room temperature, we considered black-body-induced transitions up to $n=70$. We applied a local MW field at 6.94 GHz to resonantly drive the Rydberg transition $47D_{5/2} \rightarrow 48P_{3/2}$. The radial matrix element for this Rydberg transition was calculated as $2,946.512 e a_0$, where e is the electric charge and a_0 is the Bohr radius. The corresponding dipole moment was then found as $\mu = 1,443.450 e a_0$. In detecting a MW signal, the local and signal fields were combined by a two-way MW resistive power divider (RPD) and coupled to free space via the same resonant horn antenna. The resonant coupling between hyperfine states $6S_{1/2}$, $F=4 \rightarrow 6P_{3/2}$, $F=5$ and $6P_{3/2}$, $F=5 \rightarrow 47D_{5/2}$ was realized using a 852 nm probe beam provided by a commercial extended cavity diode laser. Resonant coupling between states $6P_{3/2}$, $F=5$ and $47D_{5/2}$ was realized using a 510 nm beam generated by a frequency-doubling diode laser. The probe and coupling laser beams counterpropagated through the room-temperature Cs cell, with minimized Doppler broadening of the transition. Their polarizations were linear and parallel to the direction of the MW fields, leading to excitations of the magnetic sub-level $|m|=1/2$. For the probe beam, the $1/e^2$ beam diameter was $1.70 \pm 0.04 \text{ mm}$, and the optical power incident to the vapour cell was $120 \pm 4 \mu\text{W}$, effectively yielding $\Omega_p/(2\pi) = 5.7 \pm 0.6 \text{ MHz}$. For the coupling beam, the $1/e^2$ beam diameter was $2.00 \pm 0.05 \text{ mm}$ and the incident optical power was $34 \pm 1 \text{ mW}$, yielding $\Omega_c/(2\pi) = 0.97 \pm 0.12 \text{ MHz}$. After absorption by Cs atoms, the power of the probe light incident on the detector was $\sim 10 \mu\text{W}$. The overview of our complete experimental set-up is shown in Extended Data Fig. 1.

Note that we used the Alkali Rydberg Calculator (ARC) package to calculate the radial matrix element for $47D_{5/2} \rightarrow 48P_{3/2}$, the lifetimes of relevant atomic states, the total atom density and the Rabi frequencies of both the probe and coupling lasers.

Reduction of technical noise. The beat note results for the 852 nm laser (with an equivalent system) showed that the linewidth of the 852 nm laser was below 20 Hz and the linewidth of the 510 nm laser was estimated to be $<40 \text{ Hz}$, as a result of the following noise reductions. The frequency noise of the probe laser and the seed of the coupling laser were actively cancelled by locking them to a 10-cm-long ultralow-expansion glass cavity with a frequency noise server (FNS).

The cavity was double-coated at 852 nm and 1,020 nm with a finesse of 200,000. The linewidth of the high-finesse cavity was ~ 7.5 kHz. The cavity was placed in a vacuum system at a residual pressure below 10^{-8} mbar, and its temperature was stabilized to the zero crossing point of the coefficient of thermal expansion. The system was mounted on a passive vibration isolation platform and surrounded by an acoustic and temperature insulating box. The FNSs were realized by the Pound–Drever–Hall (PDH) technique and the feedbacks were injected to the piezoceramic (PZTs) and diode currents of both lasers. The locking bandwidths of the 852 nm laser and 1,020 nm laser were ~ 250 kHz and 350 kHz, respectively. Low-frequency intensity noise in both lasers was actively eliminated through feedback to double-pass acousto-optic modulators in a cat-eye configuration. The 852 nm probe light was separated into two orthogonally polarized output beams (using a calcite beam displacer), which propagated in parallel through the centre of the Cs vapour cell. High-frequency common-mode intensity noise in the probe light was cancelled using the balanced detection technique. Both MW sources and the digital acquisition oscilloscope were synchronized with a GPS disciplined oscillator with rubidium timebase (GPSDO) to minimize their long-time frequency drift. The signal MW source was 150 kHz detuned from the local MW source, offsetting the interference signal of the Rydberg-atom superhet to sufficiently high frequency and preventing low-frequency electronic pink noise.

Electric field measured by the Rydberg-atom superhet. Measurement of Ω_s by the atomic superhet (see equation (2) of the main text) requires knowledge of κ_s (at the optimal point) and measurement of $|P(\delta_s)|$. $|P(\delta_s)|$ was obtained by a FFT of the resonant light transmission, $P(t)$, in the presence of signal. The acquisition of κ_s involved optical spectroscopy and an adjustment by sweeping Ω_L , both circumventing the need to know Ω_s , as detailed in the following.

We have previously shown that κ can be obtained from the spectrum $\bar{P}(\Delta_c)$. Experimentally, we performed a two-peak fit of the EIT spectrum of dressed Rydberg atoms using Voigt functions $\mathcal{F}_\pm$, that is, $\bar{P}(\Delta_c) = \mathcal{F}_+(\Delta_c - \Omega_+) + \mathcal{F}_-(\Delta_c - \Omega_-)$ (Fig. 2b), and κ is readily obtained from equation (8).

Equation (9) provides a benchmark guiding us to the optimal point. There are two methods to this end. In the first method, one can measure the second derivative $\partial^2 P_0 / \partial \Omega_L^2$ as a function of Ω_L , utilizing the standard second-harmonic detection method with a lock-in amplifier. The second method measures κ while scanning Ω_L . In the first method, the optimal point corresponds to the zero crossing point $\partial^2 P_0 / \partial \Omega_L^2 = 0$, while in the second, it corresponds to the peak of $\kappa(\Omega_L)$. One can further lock the (unknown) frequency Ω_L to the zero crossing in the first scheme or the peak in the second by sending the error signal into conventional servo circuits to control the output power of the local MW source, that is, to lock to the optimal point.

We recorded $\bar{P}(\Delta_c)$ and $P(t)$ with the same digital acquisition oscilloscope. Accordingly, systematic uncertainties shared by measurements of $|P(\delta_s)|$ and $|\kappa_0|$ based on EIT spectroscopy are largely eliminated once their ratio is taken to yield Ω_s (and thus E_s). This can be seen in Extended Data Fig. 2, where we measure $|P(\delta_s)|$, $|\kappa_0|$ and E_s for the same signal, choosing different powers of control and probe lasers. In Extended Data Fig. 2a, while the measured values of $|P(\delta_s)|$ and $|\kappa_0|$ vary with the lasers, their variation patterns are generally similar. Consequently, E_s derived from their ratio only displays small fluctuations over the various laser parameters (Extended Data Fig. 2b). We note that the variation in $|P(\delta_s)|$ is not closely followed by $|\kappa_0|$ in the seventh measurement (Extended Data Fig. 2a): here, strong probe and control lasers may induce a nonlinear effect that strongly modifies the profile of the transmission spectrum.

Standard antenna method. Here, we describe how to obtain the (average) value of E_{cal} and the uncertainty of a MW electric field in the region of interaction between atoms and light based on the standard antenna method. E_{cal} is given by the standard antenna equation (IEEE Standard 1309-2013):

$$E_{\text{cal}} = \sqrt{\frac{\eta g \alpha_1 P_{\text{MW}}}{4\pi d^2}} \quad (11)$$

where $\eta = 377 \Omega$ is the intrinsic impedance of free space, P_{MW} is the power setting of the MW source, α_1 is the insertion loss between the MW source and antenna, g is the gain of the antenna and d is the distance between the transmitting antenna and the receiving point. To determine α_1 , we use a known test electric field whose strength is measured via the AT-splitting approach and substituted into equation (11). Once α_1 is known, the field strengths can be readily determined using equation (11). To obtain the uncertainty in the generation of E_{cal} , we use the calibration information of the pre-calibrated MW source (R&S SMF100A microwave signal generator, Rohde & Schwarz). When the setting power P_{MW} of both sources is above -10 dBm, the uncertainty is 0.8 dB. When -70 dBm $< P_{\text{MW}} < -10$ dBm, the uncertainty is 0.9 dB, and when -90 dBm $< P_{\text{MW}} < -70$ dBm, the uncertainty is 1 dB. In Fig. 3a, we used both the average value (the dashed line) and uncertainty (the light blue region) from the standard antenna method.

In the proof-of-principle demonstrations shown in Fig. 2, we used the average value of electric field from the standard antenna method as the true value of the electric field for both the signal and local MW. Because they are emitted

from difference sources, each required calibration. For the local MW field, the insertion loss between the MW source and antenna was found to be $\alpha_1 = 14.7$ dB. This includes 6.5 dB RPD insertion loss and 4 m of 1.5 dB m^{-1} wire loss, while the remaining insertion loss can be attributed to connector insertion loss. For the signal field, we obtained $\alpha_1 = 11.9$ dB, which is smaller than for the local MW due to use of a shorter transmission wire (3 m).

Sensitivity spectrum and noise analysis. We experimentally obtained the sensitivity spectrum (sensitivity as a function of signal frequency) associated with the electric field measurement by the Rydberg-atom superhet (Extended Data Fig. 3). Specifically, we measured the probe laser transmission under normal resonant conditions in the absence of signal MW (we denote by $|P(\delta_s)|$) the corresponding single-sided Fourier spectrum, for unit RBW). Following from equation (2), we derive the sensitivity spectrum $S(\delta_s) = [\hbar/(\sqrt{2}\mu)]|P(\delta_s)|/|\kappa_0|$ for the measured electric field.

Extended Data Fig. 3 reveals the primary source of noise limiting the sensitivity of the Rydberg-atom superhet, which varies depending on the signal frequency δ_s . For frequencies below 1 kHz, the $1/f$ noise of electric circuits dominates over other noises. For frequencies between 1 kHz and 100 kHz, the transit noise due to thermal atoms provides the main noise source. For frequencies above 100 kHz, it is the frequency noise of the coupling and probe lasers caused by FNS resonance that mainly limits the sensitivity. Given that the noise spectrum with and without the local MW does not exhibit qualitative variation, we conclude that the local MW field does not provide a main noise source for our measured electric field. In our experiments, we chose $\delta_s = 150$ kHz in view of the requirements of both optimal sensitivity and $2\pi\delta_s \ll \Gamma_{\text{EIT}}$. The corresponding value of $S(\delta_s)$, determined by the frequency noise, provides the noise base of our electric field measurement shown in Fig. 3a.

Principal error sources of uncertainty. From the noise spectrum associated with the electric field measurements by the atomic superhet, we see that the minimum absolute uncertainty is 55×10^{-9} V cm^{-1} , determined by the noise base. Here, we analyse the main sources of systematic and statistical error that contribute to the 1σ relative standard measurement error σ_s associated with Ω_s measurements in a 1 s measurement time based on equation (2). Let the 1σ relative standard measurement error associated with measurements of $|P(\delta_s)|$ and $|\kappa_0|$ be denoted by σ_p and σ_κ , respectively. We have $\sigma_s = \sqrt{\sigma_p^2 + \sigma_\kappa^2}$. For σ_p in the measurement of $|P(\delta_s)|$, the main relative statistical uncertainty is $1/2^8$, determined by the 8 bit vertical resolution of our digital oscilloscope (R&S RTO2000 oscilloscope, Rohde & Schwarz) when $E_s \gg 8 \mu\text{V cm}^{-1}$, whereas for $E_s \ll 8 \mu\text{V cm}^{-1}$ it is determined by $1/\text{SNR}$. Here, $\text{SNR} = |P(\delta_s, \Omega_s)|/\bar{P}(\delta_s)$ for the optical readout. The relative systematic uncertainty is always given by $\sqrt{1 + 1/\text{SNR}} - 1$, which is much smaller than $1/2^8$ when $E_s \gg 8 \mu\text{V cm}^{-1}$, but becomes comparable to the relative statistical uncertainty for $E_s \ll 8 \mu\text{V cm}^{-1}$. Therefore, σ_p is dominated by $1/2^8$ for $E_s \gg 8 \mu\text{V cm}^{-1}$, but is controlled by SNR for $E_s \ll 8 \mu\text{V cm}^{-1}$. For σ_κ in the measurement of $|\kappa_0|$, because $|\kappa_0|$ is measured from the probe laser transmission spectrum in the presence of a strong local MW (for $\Omega_s = 0$), σ_κ is mainly determined by the statistical relative uncertainty of $1/2^8$ from the vertical resolution of the digital oscilloscope. Note that the aforementioned statistical relative uncertainty can be significantly reduced by using a digital oscilloscope with higher vertical resolution. The present uncertainty analysis is consistent with our experiment in Fig. 3a.

Measuring the phase from the EIT signal. We extract a phase ϕ_{out} from $P(t)$ using a lock-in amplifier. The filter slope of the lock-in amplifier was set to 18 dB oct^{-1} and the time constant was fixed at 100 ms to realize 1.04 Hz equivalent noise bandwidth. This yields a SNR of 44 dB for $E_s = 7.8 \mu\text{V cm}^{-1}$, leading to theoretical estimation of the phase resolution as 0.6° . To assess the phase resolution experimentally, the 1σ standard error of fluctuations in ϕ_{out} was measured in a period of 1–5 s and 6–10 s, respectively, with 1 s waiting time for the signal to reach 99% of its final value.

Roadmap to QPNL sensitivity. Here, we determine the QPNL sensitivity for our set-up, which is formally given by $E_{\text{QPNL}} = \frac{\hbar}{\sqrt{2}\mu} \frac{1}{\sqrt{N_s \tau_c}}$. Here $\tau_c = \Gamma_{\text{sp}}^{-1}$ is the coherence time of the Rydberg-atom superhet discussed earlier, and N_s is the number of atoms participating in the EIT process per second. For our set-up, we estimate τ_c from Γ_{EIT}^{-1} with $\Gamma_{\text{EIT}}/(2\pi) = 7.9$ MHz from experiment results. N_s is estimated as $2.14 \times 10^{13} \text{ s}^{-1}$ for the EIT process, which leads to an enhancement of $5 \mu\text{W}$ light transmission at the photon detector compared to that without the coupling laser. This gives $E_{\text{QPNL}} \approx 700 \text{ pV cm}^{-1} \text{ Hz}^{-1/2}$, as indicated in Extended Data Fig. 3.

Extended Data Fig. 3 further suggests a roadmap towards obtaining QPNL sensitivity. For $\delta_s = 150$ kHz, the primary noise sources in our scheme can be systematically eliminated using techniques that are feasible in current quantum sensing experiments: (1) the transit noise due to thermal atoms can be eliminated using larger-diameter probe and coupling beams; (2) the laser frequency noise can be eliminated by using state-of-the-art lasers with mHz linewidth^{31,32} and expanding the servo bandwidth to several MHz^{33,34}; (3) the amplifier noise of the photon detector can be reduced through optical heterodyne or homodyne detection; (4) the spectrum analyser noise can be removed by using conventional

electronic amplifiers. After eliminating relevant technical noise, the Rydberg-atom superhet approaches QPNL sensitivity.

Data availability

The data represented in Figs. 2, 3 and 4 and Extended Data Figs. 2 and 3 are available as Source Data. All other data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

L.X., S.J., L.Z. and M.J. proposed the project. M.J. and Y.H. developed the research. M.J. and L.Z. performed the experiments and analysed the experimental data. Y.H. and M.J. developed the theory and carried out theoretical calculations. M.J., L.Z. and H.Z. contributed to the experimental set-up. Y.H., M.J., L.X., J.M. and L.Z. wrote the manuscript and provided revisions. All authors contributed to discussions of the results and the manuscript.

Competing interests

The authors declare no competing interests.

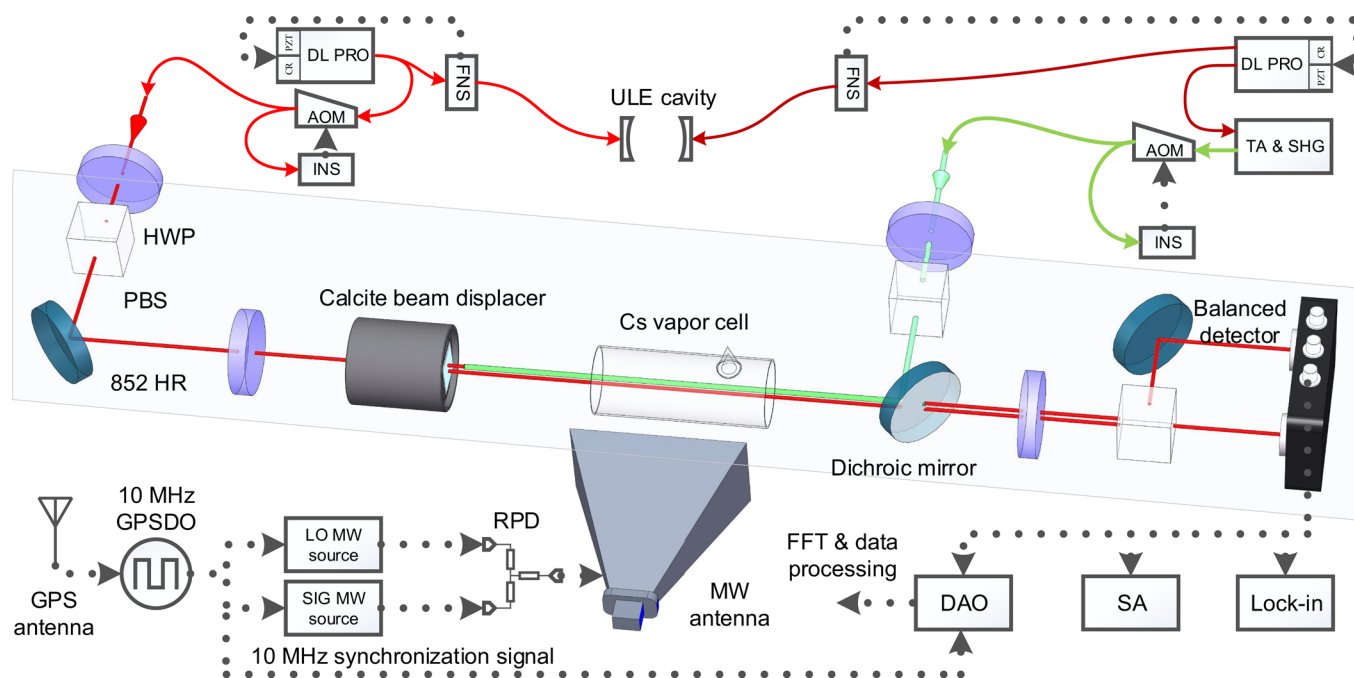
Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41567-020-0918-5>.

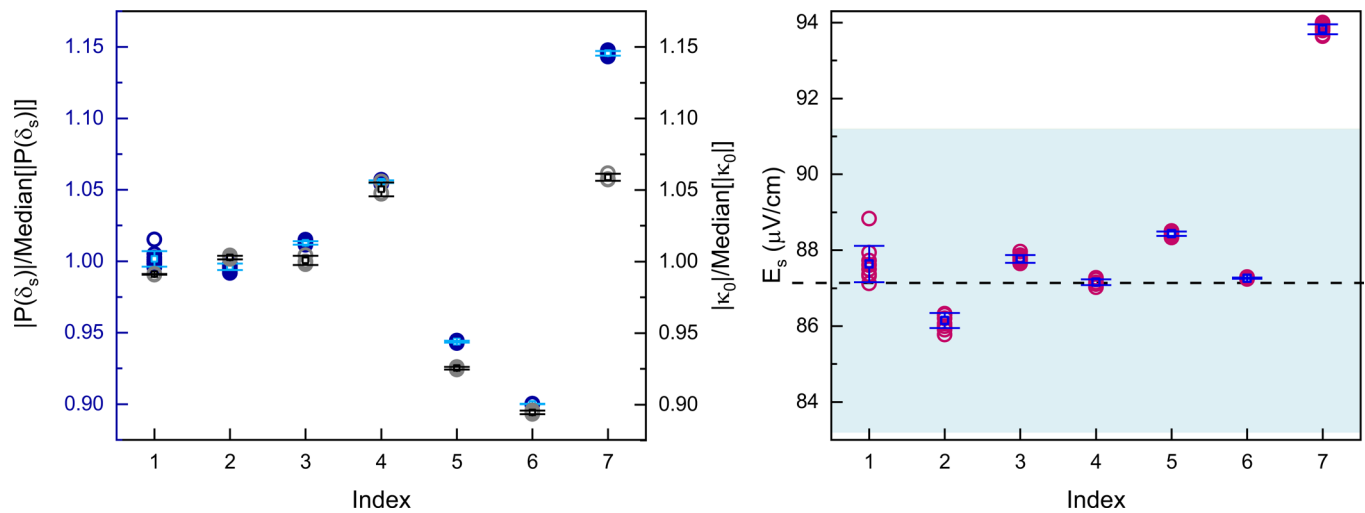
Supplementary information is available for this paper at <https://doi.org/10.1038/s41567-020-0918-5>.

Correspondence and requests for materials should be addressed to L.Z. or L.X.

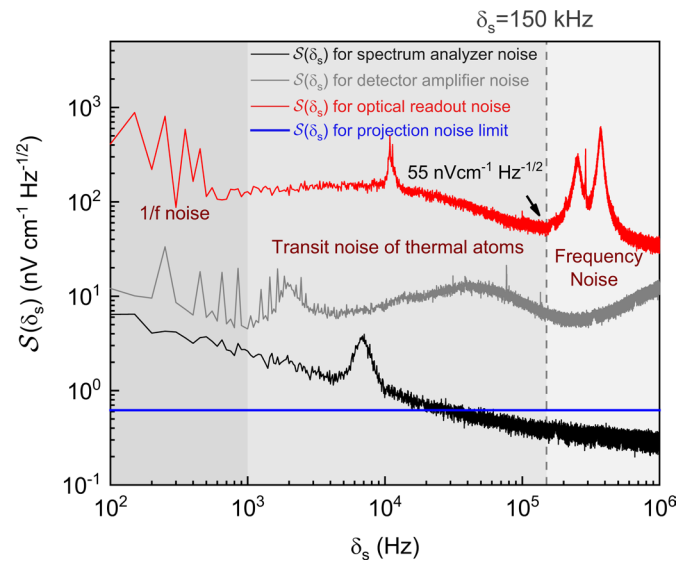
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Extended Data Fig. 1 | Overview of the experimental setup. We have used the following notations: (1) DL PRO: external-cavity diode lasers, (2) CR: diode laser current, (3) PZT: piezo ceramic, (4) AOM: The double-pass acousto-optic modulators which shift the frequencies of the probe and coupling lights to atomic resonances, (5) INS: intensity noise server, (6) FNS: frequency noise server, (7) ULE: ultra-low expansion (ULE) glass cavity, (8) TA & SHG: tapered amplifier and second-harmonic generation, (9) HWP: half-wave plate, (10) PBS: polarizing beam splitter, (11) HR: dielectric mirror, (12) GPSDO: GPS disciplined oscillator with Rubidium timebase, (13) RPD: 2-way microwave resistive power divider, (14) FFT: fast Fourier transform, (15) DAO: digital acquisition oscilloscope, (16) SA: spectrum analyzer, (17) Lock-in: lock-in amplifier.



Extended Data Fig. 2 | Measurements of the same signal for various laser powers. In measuring the same signal, seven different choices of (P_p, P_c) for the powers of probe and control lasers are considered, indexed by $i = 1, \dots, 7$ in the horizontal axis of **a** and **b**. They are respectively $(120 \mu\text{W}, 34 \text{ mW})$, $(108 \mu\text{W}, 30 \text{ mW})$, $(120 \mu\text{W}, 30 \text{ mW})$, $(132 \mu\text{W}, 34 \text{ mW})$, $(120 \mu\text{W}, 27 \text{ mW})$, $(84 \mu\text{W}, 30 \text{ mW})$, $(180 \mu\text{W}, 39 \text{ mW})$. Panel **a** presents measurements of $|P(\delta_s)|$ and $|\kappa_0|$ by atomic superhet, and panel **b** shows $E_s = [\hbar/(\sqrt{2} \mu)]|P(\delta_s)|/|\kappa_0|$. In panel **a**, the dark blue circle denotes single-shot measurement of $|P(\delta_s)|$. The light blue square depicts average $|P(\delta_s)|$ from 10 single-shot measurements with error bar indicating the 1σ standard error. All data of $|P(\delta_s)|$ has been normalized by the median of $|P(\delta_s)|$. The gray circle denotes single-shot measurement of $|\kappa_0|$. Average $|\kappa_0|$ from 3 measurements is shown (black square) with error bar denoting 1σ standard error. Data of $|\kappa_0|$ has been normalized by its median. In panel **b**, the red circle represents single-shot measurement of E_s by atomic superhet, and the blue square depicts average E_s from 10 measurements with error bar indicating the 1σ standard error. The same signal electric field is also measured using standard antenna method (dashed black line), and the light blue patch denotes measurement uncertainty in this method.



Extended Data Fig. 3 | Sensitivity spectrum of the Rydberg-atom superhet. Measured sensitivity S as a function of relative signal frequency δ_s is shown for various optical readout noises (red curve), the amplifier noise of photon detector (gray curve) and the spectrum analyzer noise (black curve). These optical readout noises include 1/f noise, transit noise of atoms, and laser frequency noise. The blue line represents the QPNL sensitivity of our setup (Methods). The dashed gray line indicates $\delta_s = 150$ kHz chosen in experiments in the main text. The sensitivity of our setup is limited by laser frequency noise to $55 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$.